

Design and Implementation of an IoT-Based Real-Time Gas Pipeline Leakage Detection and Automated Shut-Off System

Akshay S^{1*}, Harinarayanan P¹, Raj K Panicker¹, S Sreesharan¹, Rejin S², Gijoy S²

¹B.Tech Student, Department of Mechanical Engineering, College of Engineering Trivandrum, Thiruvananthapuram, Kerala, India

²Assistant Professor, Department of Mechanical Engineering, College of Engineering Trivandrum, Thiruvananthapuram, Kerala, India

*Corresponding Author Email: ssreesharan07@gmail.com

Abstract—Gas leakage in residential and industrial pipeline networks presents a critical safety hazard due to its potential to cause explosions, fire accidents, and environmental damage. Rapid detection and immediate containment are therefore essential to prevent catastrophic incidents. This study presents the design and implementation of an IoT-based real-time gas pipeline safety system capable of detecting gas leaks and automatically initiating protective actions. The proposed system integrates MQ-series gas sensors with an ESP32 microcontroller for continuous monitoring of gas concentration levels. Sensor data are transmitted through the MQTT publish-subscribe protocol to a Node.js-based backend server that processes real-time measurements and triggers automated responses when hazardous thresholds are exceeded. Upon leak detection, the system immediately activates a solenoid valve to shut off gas supply, while simultaneously issuing alerts through a web-based monitoring dashboard. The system architecture also incorporates a lightweight analytical layer capable of generating real-time safety insights to assist operators during emergency conditions. Experimental evaluation under simulated leakage scenarios demonstrated rapid detection and response, with automated valve closure occurring within approximately 500 ms, significantly reducing reaction time compared to manual intervention. The proposed architecture provides a low-latency, scalable and cost-effective safety solution suitable for both residential and industrial gas pipeline monitoring applications.

Keywords—IoT, Gas Leak Detection, ESP32, MQTT, Pipeline Safety, Automated Shut-Off System

I. INTRODUCTION

The growing demand for energy across residential, commercial, and industrial sectors has resulted in the extensive use of gas-based energy systems such as liquefied petroleum gas (LPG) and natural gas pipeline networks. These systems are widely adopted due to their efficiency, cost-effectiveness, and relatively lower environmental impact compared to other fossil fuels. Despite these advantages, gas leakage remains a critical safety issue in gas distribution

infrastructure. Leakage incidents can lead to severe consequences including explosions, fire hazards, and environmental contamination. Such failures may arise due to equipment malfunction, pipeline corrosion, faulty regulators, improper installation, or operational negligence. These incidents pose significant threats to human life, property, and industrial facilities, emphasizing the need for reliable and rapid gas leakage detection mechanisms. Consequently, the development of intelligent gas monitoring and safety systems has attracted increasing attention in recent years as a means to improve operational safety and minimize risks associated with gas distribution networks.

Recent research has focused on sensor-based monitoring systems capable of detecting gas leakage and providing early warning alerts. Bhalawe et al. [1] proposed an Internet of Things (IoT)-enabled gas detection system that integrates gas sensors, microcontrollers, and wireless communication modules to monitor gas leakage and notify users through mobile applications. Their study demonstrated the feasibility of remote monitoring and real-time alert mechanisms for enhancing safety in domestic environments. However, the system primarily focuses on detection and notification and does not incorporate automated control mechanisms to prevent further gas leakage once a hazard is detected.

Further efforts have been made to integrate gas leakage detection with automated safety response mechanisms. Bairagi et al. [2] developed an intelligent safety system capable of detecting LPG leakage and automatically shutting off the gas supply using a microcontroller-controlled valve mechanism. The system also includes alarms and SMS notification features to alert users of potential hazards. Experimental evaluations indicated that the system could successfully detect gas leakage and interrupt gas flow within a few seconds, thereby reducing accident risks. Nevertheless, the proposed solution is mainly designed for domestic LPG cylinder applications and does not fully address the requirements of large-scale gas pipeline monitoring systems.

In addition to hardware-based monitoring solutions, IoT-enabled safety frameworks have also been explored for improving gas leakage detection and response. Faye et al. [3] introduced a smart gas leakage detection system that integrates an MQ-6 gas sensor, an Arduino-based microcontroller, and a servo motor to detect gas leaks and automatically shut off gas supply. The system also generates audible alerts using buzzers and display modules. Although this design offers a cost-effective and automated safety approach, it mainly focuses on localized gas detection systems and lacks advanced data processing capabilities for large-scale monitoring applications.

More recently, advanced approaches have incorporated artificial intelligence and multimodal sensing techniques to enhance detection accuracy. Attallah [4] proposed a deep learning-based gas leakage detection framework that combines electronic nose sensor data with infrared thermal

imaging. The system employs convolutional neural networks for spatial feature extraction and bidirectional long short-term memory networks for temporal analysis, enabling accurate gas leakage detection through multimodal data fusion. While the proposed approach demonstrates promising results, it requires significant computational resources and carefully controlled datasets, which may limit its practical implementation in real-time industrial environments.

Although considerable progress has been achieved in the development of gas leakage detection technologies, existing solutions often focus either on simple sensor-based detection systems without automated safety responses or on computationally intensive models that may not be feasible for practical deployment. Furthermore, many current systems are designed primarily for domestic LPG cylinder monitoring rather than for continuous monitoring of gas pipeline infrastructure. Therefore, there remains a need for a scalable, reliable, and cost-effective monitoring system capable of detecting gas leakage, initiating automated safety responses, and providing real-time monitoring through connected platforms.

Motivated by these challenges, the present study proposes an IoT-based intelligent gas pipeline safety system designed to enable continuous monitoring, rapid leakage detection, and automated safety response mechanisms. The proposed system integrates gas sensors, microcontroller-based processing units, and wireless communication technologies within an IoT framework to monitor gas concentration levels in real time. When abnormal gas levels are detected, the system automatically triggers alert mechanisms and activates safety control actions such as shutting off the gas supply to prevent further leakage. The proposed approach aims to enhance pipeline safety, reduce response time during leakage incidents, and provide a practical monitoring solution for both residential and industrial gas distribution systems.

II. MATERIALS AND METHODS

A. Simulation-Based Virtual Prototyping

The initial stage of system development involved simulation-based virtual prototyping to evaluate the proposed gas leakage detection and safety control framework before physical implementation. A digital twin environment was created using the Wokwi IoT simulator, which enables accurate simulation of embedded hardware systems and cloud connectivity in a controlled environment.

Within this virtual environment, the core hardware components of the proposed system—including the ESP32 microcontroller, MQ-2 gas sensors, relay modules, and alarm components—were simulated to replicate real-world operating conditions. This approach allowed the research team to model potential gas leakage scenarios and evaluate

system behavior without the risks associated with testing with actual gas sources.

To control the simulation environment, the firmware logic for the ESP32 microcontroller was developed using C++ programming language. The implemented control code handled sensor data acquisition, threshold-based leak detection, and execution of safety actions such as triggering alarms and activating shut-off mechanisms.

The virtual prototyping stage also enabled Hardware-in-the-Loop (HIL) style testing, where the embedded control logic was validated against simulated sensor inputs. This process ensured that the automated safety logic—including leak detection thresholds and valve shut-off commands—operated correctly under different simulated conditions. By verifying the system behavior in a risk-free digital environment, potential errors in the firmware and control logic could be identified and corrected before deploying the system on physical hardware.

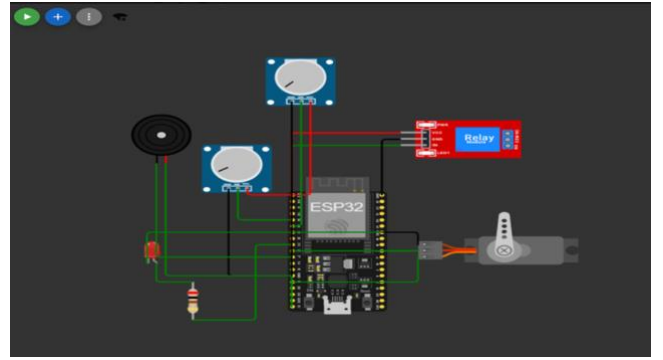


Fig. 1. Simulation-based virtual prototyping of the IoT gas leakage detection system implemented in the Wokwi simulator, showing ESP32 microcontroller, MQ-2 gas sensors, relay module, servo actuator, and alarm components.

B. Data Transport and Communication Architecture

The communication framework responsible for transmitting real-time data between distributed sensing units and the central monitoring server is implemented through a lightweight data transport architecture. In the proposed system, the Message Queuing Telemetry Transport (MQTT) protocol is employed to facilitate reliable communication between the embedded hardware and the cloud-based backend infrastructure. MQTT operates on a publish-subscribe communication model, where sensor nodes publish data to specific topics while subscribed clients receive the corresponding messages through a broker service.

A HiveMQ MQTT broker was used to manage message distribution between the ESP32-based sensing nodes and the backend processing server. In this configuration, gas concentration data collected by the sensors are continuously published to designated MQTT topics, which are subsequently received and processed by the monitoring

server. Similarly, control commands generated by the server—such as valve shut-off instructions—are transmitted back to the embedded system through subscribed control topics.

The selection of MQTT for the proposed architecture is primarily motivated by its low-latency and bandwidth-efficient communication capabilities, which are particularly suitable for Internet of Things (IoT) applications. Unlike traditional request–response communication protocols such as HTTP, MQTT enables event-driven message delivery, where data are instantly pushed to subscribed clients without requiring continuous polling from the server. This mechanism significantly reduces communication overhead and ensures rapid delivery of safety-critical information.

Such real-time and bidirectional communication capability is essential for safety monitoring systems, where immediate response to hazardous conditions is required. By utilizing MQTT-based communication, the proposed system enables fast transmission of gas leakage alerts and rapid execution of automated control actions, thereby minimizing response time and improving overall pipeline safety.

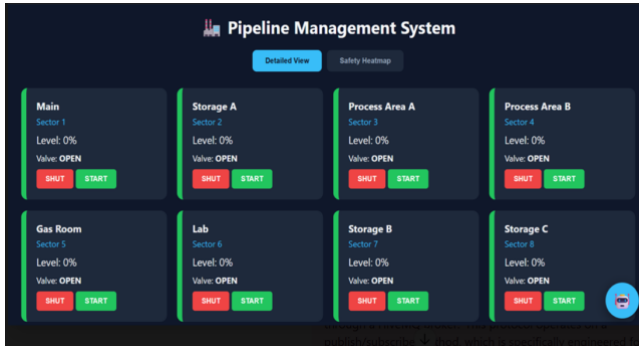


Fig. 2. Real-time monitoring dashboard of the IoT-based pipeline safety system showing gas concentration levels and valve status across multiple pipeline sectors.

C. Real-Time Monitoring Dashboard and Safety Heatmap Visualization

To enable continuous monitoring and rapid situational awareness, a web-based Human–Machine Interface (HMI) dashboard was developed to visualize real-time gas concentration data across multiple pipeline sectors. The dashboard provides operators with an interactive interface displaying the operational status of different monitoring zones within the pipeline network.

A heatmap-based visualization module was implemented to represent gas concentration levels across different sectors of the facility. Each sector is displayed as a color-coded tile where the color intensity reflects the detected gas concentration level. Low gas concentrations are represented by green indicators, while elevated concentrations transition toward orange and red, indicating potential safety hazards.

The heatmap visualization allows operators to quickly identify critical leakage zones without manually analyzing numerical sensor readings. The dashboard receives real-time sensor updates through the MQTT communication framework and dynamically updates the heatmap display, ensuring immediate visualization of abnormal gas conditions. This visualization approach enhances situational awareness, fault localization, and decision-making efficiency, enabling operators to rapidly identify hazardous regions and initiate corrective actions when necessary.

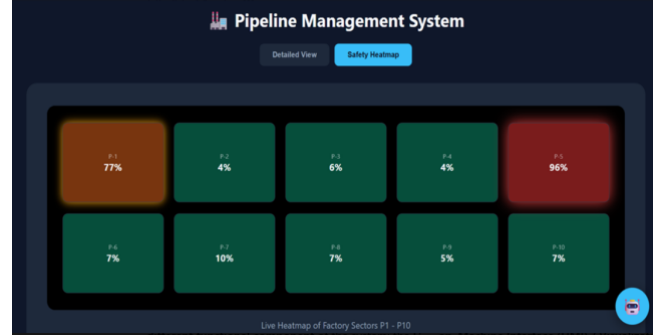


Fig. 3. Real-time safety heatmap visualization of gas concentration levels across multiple pipeline sectors (P1–P10) displayed on the monitoring dashboard. Color intensity represents the severity of detected gas levels.

D. Backend Processing and Control Architecture

The backend processing unit functions as the central control layer of the proposed pipeline monitoring system, responsible for receiving sensor data, analyzing gas concentration levels, and issuing control commands to the embedded hardware. The server-side infrastructure was developed using a Node.js v25 environment operating in ECMAScript Module (ESM) mode, integrated with the Express.js framework to manage application routing and data handling.

Node.js was selected due to its asynchronous and non-blocking event-driven architecture, which is well suited for real-time Internet of Things (IoT) applications where multiple devices continuously transmit data. Incoming sensor readings published through the MQTT communication layer are processed by the backend server, where gas concentration values are evaluated against predefined safety thresholds.

The implemented backend architecture ensures high scalability and reliability under dynamic operating conditions. By utilizing asynchronous request handling, the system is capable of managing data streams from multiple sensing nodes simultaneously without performance degradation. In the current implementation, the server is designed to handle monitoring operations for up to ten independent pipeline sectors, enabling concurrent detection and mitigation of multiple gas leakage events. Upon detecting abnormal gas

concentration levels, the backend processing unit immediately generates control instructions and publishes appropriate commands to the corresponding MQTT control topics. These commands trigger safety mechanisms such as automatic valve closure and alarm activation, thereby enabling rapid response to hazardous conditions and ensuring the safe operation of the gas distribution system.

E. AI-Assisted Safety Analysis and Decision Support

To enhance the analytical capability of the proposed pipeline monitoring system, an artificial intelligence (AI)-assisted analysis layer was integrated into the backend processing architecture. This module utilizes the Google Gemini 1.5 Flash AI model to interpret system events and generate contextual safety insights during abnormal operating conditions.

The AI layer functions as an intelligent supervisory module that analyzes real-time sensor data and system alerts received from the monitoring infrastructure. Unlike conventional threshold-based safety systems that rely solely on binary decision logic, the AI module provides higher-level analytical interpretation of detected gas leakage events. By processing contextual information related to gas concentration levels, affected pipeline sectors, and system status, the AI system generates human-readable safety reports and response recommendations.

During a gas leakage event, the AI model evaluates the severity and spatial distribution of the detected hazard and produces structured guidance for system operators. These insights may include identification of high-risk sectors, recommended safety actions, and situational summaries to assist emergency response teams. By incorporating AI-assisted analytical capabilities, the proposed system enhances situational awareness, operational transparency, and decision-making support for operators responsible for managing gas distribution infrastructure.

F. Human–Machine Interface (HMI) for Real-Time Visual Monitoring

A dedicated Human–Machine Interface (HMI) was developed to enable continuous visual monitoring of the pipeline safety system by operators. The HMI is implemented as a web-based dashboard that provides real-time visualization of gas concentration levels, pipeline sector status, and safety alerts across the monitored infrastructure.

The dashboard interface was developed using standard web technologies including HTML, CSS, and JavaScript, ensuring compatibility across multiple devices and platforms. To support real-time data streaming between the backend server and the user interface, the system utilizes the Socket.io communication library, which enables WebSocket-based connectivity. Through WebSocket synchronization, the

dashboard maintains continuous bi-directional communication with the backend processing server. This architecture allows sensor updates, system alerts, and control actions to be transmitted instantly to the user interface without requiring manual page refreshes. As a result, operators can observe real-time system behavior, monitor gas concentration levels across different pipeline sectors, and respond promptly to abnormal operating conditions. The integration of this interactive monitoring interface significantly enhances operator situational awareness, enabling faster detection of hazardous events and facilitating timely intervention in gas leakage scenarios.

G. Automated Safety Control Logic

To ensure rapid mitigation of hazardous conditions, the proposed system incorporates an automated safety control mechanism designed to detect gas leakage events and initiate immediate protective actions. The control logic operates through a continuous monitoring framework in which gas concentration levels measured by the sensing units are evaluated in real time by the backend processing system.

Gas concentration values are normalized on a proportional scale ranging from 0 to 100%, representing the relative intensity of detected gas levels within each monitored pipeline sector. A predefined safety threshold is established to determine hazardous conditions. When the measured gas concentration reaches or exceeds 90% of the defined threshold, the system identifies the situation as a critical gas leakage event.

Upon detection of this threshold breach, the backend server immediately generates a control command and publishes a "CLOSE" instruction to the corresponding MQTT control topic associated with the affected pipeline segment. This command is transmitted through the MQTT communication framework to the embedded control unit responsible for hardware actuation.

Upon receiving the control instruction, the embedded system activates the solenoid valve mechanism, which rapidly shuts off the gas supply to the compromised pipeline section. This automated response mechanism eliminates the delay associated with manual intervention and significantly reduces the risk of explosion or fire caused by uncontrolled gas leakage. Through this coordinated sequence of continuous monitoring, threshold detection, server-side decision making, and hardware actuation, the system ensures rapid containment of gas leakage events and enhances overall operational safety in gas pipeline environments.

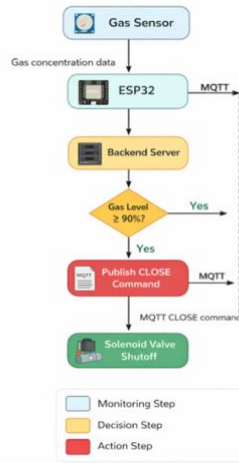


Fig. 4. Automated safety control workflow for gas leakage detection and mitigation.

H. System Runtime Monitoring and Debugging

To ensure reliable system operation and facilitate effective debugging during development and testing, a runtime monitoring framework was implemented for observing the behavior of the backend infrastructure and communication processes. The monitoring environment was established using Windows PowerShell, which served as the primary interface for tracking system activity during operation.

Through this interface, developers and system operators were able to monitor live API requests, backend server logs, and MQTT communication traffic in real time. This capability enabled continuous verification of data exchange between sensing nodes, the backend server, and the monitoring dashboard. Runtime monitoring also allowed the identification of potential communication delays, system exceptions, or abnormal operational behavior within the IoT framework. By observing real-time server responses and message flows, the stability of the sensor-to-server-to-actuator communication pipeline could be validated under active operating conditions. The integration of runtime monitoring tools contributed to improved system reliability, debugging efficiency, and operational transparency, ensuring that the complete pipeline monitoring and safety control architecture functioned correctly during experimental testing.

I. Basic working prototype

To validate the practical functionality of the proposed gas leakage detection and automated shut-off system, a hardware prototype was developed using an embedded microcontroller platform and gas sensing components. The prototype system consists of an ESP32 microcontroller, an MQ-2 gas sensor module, a relay module, and a solenoid valve actuator that represents the gas shut-off mechanism. In addition, a web-

based monitoring dashboard was implemented to visualize real-time system status and gas concentration levels.

The MQ-2 gas sensor was used as the primary sensing element for detecting combustible gases such as liquefied petroleum gas (LPG), methane, and propane. The sensor continuously measures gas concentration in the surrounding environment and converts the detected gas intensity into an analog voltage signal. This signal is transmitted to the ESP32 microcontroller for processing and analysis.

The ESP32 microcontroller functions as the central control unit of the system. It continuously reads the sensor data and compares the measured gas concentration with a predefined safety threshold. Under normal conditions, when the gas level remains below the threshold value, the system maintains normal operation and the gas supply remains active.

When the gas concentration exceeds the predefined threshold level, the ESP32 immediately triggers the safety control mechanism. A digital control signal is sent to the relay module, which acts as an electrical interface between the microcontroller and the solenoid valve. Once activated, the relay energizes the solenoid valve, causing it to close and thereby automatically shutting off the gas supply to prevent further leakage.

In addition to the automated safety response, the system transmits real-time sensor data to a web-based monitoring dashboard through the communication framework implemented in the project. The dashboard displays gas concentration levels, system status, and valve conditions, allowing operators or users to visually monitor the system behavior in real time. When a gas leakage event occurs, the dashboard immediately reflects the abnormal gas level and indicates that the valve has been automatically closed.

Experimental testing of the prototype was conducted under controlled conditions by exposing the MQ-2 sensor to simulated gas leakage scenarios. During these tests, the system successfully detected elevated gas concentrations, automatically activated the shut-off mechanism, and simultaneously updated the monitoring dashboard with the corresponding system status. The results demonstrate that the proposed system is capable of providing both rapid automated safety response and real-time monitoring, thereby improving operational safety and situational awareness.

The successful operation of the prototype confirms that the integration of gas sensing, embedded control, automated shut-off mechanisms, and real-time dashboard monitoring provides an effective and reliable solution for gas leakage detection and mitigation in both residential and industrial gas pipeline environments.

To enhance the usability of the proposed gas monitoring system, an AI-based chatbot module was integrated into the monitoring dashboard to assist users in understanding system status and responding to emergency situations. The chatbot analyzes real-time sensor data and system alerts received from the monitoring infrastructure and provides contextual safety information to the user.

When abnormal gas concentration levels are detected, the chatbot generates automated responses describing the detected issue, affected pipeline sector, and recommended safety actions. This allows operators to quickly interpret system alerts without manually analyzing raw sensor data. In addition, users can interact with the chatbot through the dashboard interface to obtain information about system status, gas concentration levels, and safety recommendations.

The integration of the AI chatbot improves user interaction with the monitoring system and enhances situational awareness during gas leakage events. By providing real-time explanations and safety guidance, the chatbot supports faster decision-making and contributes to improved operational safety.

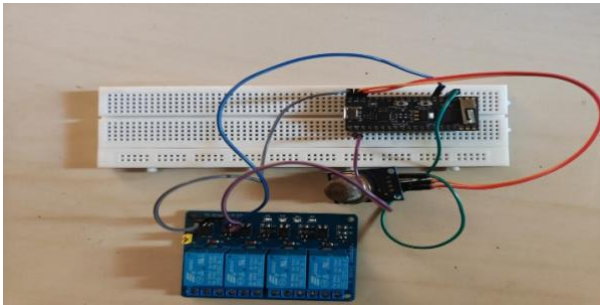


Fig.5. Basic working prototype

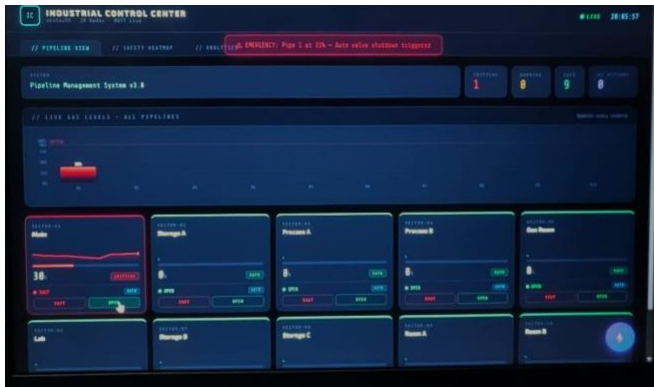


Fig .6. Automatic shutdown completed

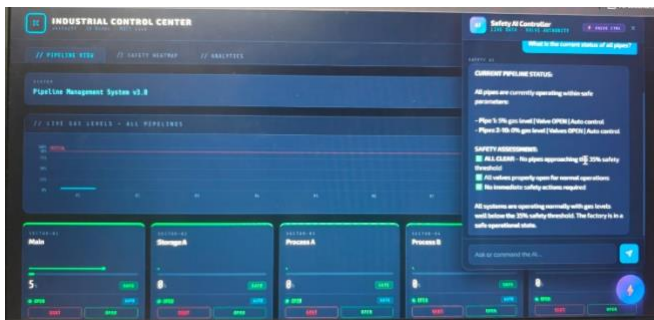


Fig .7. Safety AI controller.

III. RESULTS AND DISCUSSION

A. System Responsiveness

The responsiveness of the proposed IoT-based gas pipeline safety system was evaluated under simulated leakage conditions to determine its ability to detect hazards and initiate automated safety actions. Experimental observations indicate that the system responds significantly faster than manual operator intervention.

When the gas concentration exceeded the predefined safety threshold, the automated shut-off mechanism was activated within approximately 500 ms. At the same time, the WebSocket-based communication framework enabled real-time synchronization between the backend server and the monitoring dashboard. The dashboard interface was updated in less than 100 ms, allowing operators to observe hazardous conditions almost instantly.

The AI-assisted analysis layer also demonstrated efficient processing performance. Upon detection of abnormal gas levels, the system generated a structured safety summary and situational assessment within 1–2 seconds, providing additional decision-support information for operators. Furthermore, the integrated heatmap visualization successfully highlighted the location and intensity of gas leakage across monitored pipeline sectors, thereby improving situational awareness and facilitating rapid identification of high-risk zones.

B. Reliability and Safety Impact

The reliability of the proposed system was assessed through multiple simulated gas leakage scenarios. Under these conditions, the automated safety control mechanism consistently performed as intended, successfully triggering valve shut-off actions whenever hazardous gas levels were detected.

The system achieved a 100% successful valve closure rate during all simulated leak events once the gas concentration exceeded the predefined threshold. The MQ-series gas sensors exhibited reliable detection performance, with early warning signals being triggered at gas concentrations above 400 ppm, enabling timely preventive action before reaching critical hazard levels.

One of the most significant safety advantages of the proposed system is the elimination of delays associated with manual intervention. Traditional safety systems typically rely on operator response after receiving an alert notification. In contrast, the proposed architecture implements automated containment, immediately isolating the affected pipeline segment upon detection of a leak.

This shift from reactive alert-based safety mechanisms to proactive automated mitigation significantly enhances the overall safety of gas pipeline operations. By combining real-time sensing, rapid communication through MQTT, automated control logic, and interactive visualization through the HMI dashboard, the system provides a comprehensive safety monitoring solution capable of reducing the risk of accidents in both residential and industrial environments.

IV. CONCLUSION

This study presented the design and implementation of an IoT-based intelligent gas pipeline safety system aimed at improving the detection and mitigation of gas leakage hazards. The proposed architecture integrates gas sensors, an ESP32 microcontroller, MQTT-based communication, a Node.js backend processing framework, and a web-based human-machine interface to enable real-time monitoring and automated safety control. The system continuously evaluates gas concentration levels and automatically initiates safety actions when hazardous conditions are detected.

Experimental evaluation demonstrated that the system is capable of detecting gas leakage events and triggering automated valve shut-off within approximately 500 milliseconds, significantly reducing response time compared to manual intervention. The real-time dashboard and heat map visualization further enhanced operator situational awareness by clearly identifying the location and intensity of gas leaks. In addition, the integration of an AI-assisted analysis layer provided contextual safety insights to support decision-making during emergency situations.

Overall, the proposed system transforms conventional gas safety monitoring from a reactive alert-based approach to a proactive automated containment mechanism, thereby improving operational safety in both residential and industrial gas distribution environments. Future work will focus on integrating predictive analytics for early leak forecasting and expanding the system to support large-scale pipeline monitoring using distributed sensor networks.

V. REFERENCES

- [1] Bhalawe, L., Marotkar, S., Dayande, Y., & Chauhan, H. S. (2024). IoT based gas detection and notifying system. *International Journal of Research Publication and Reviews*, 5(6), 2892–2896. <https://doi.org/10.55248/gengpi.5.0624.1514>
- [2] Bairagi, P. P., Patir, B., Bora, P., Deka, L., & Rajbongshi, S. K. (2024). SmartGuard: An intelligent LPG leakage detection and control system. *International Journal of Engineering Trends and Technology*, 72(9), 265–272. <https://doi.org/10.14445/22315381/IJETT-V72I9P121>
- [3] Faye, P., Kuhikar, S., Rathod, R., Chandewar, G., & Thakre, K. (2025). Smart IoT based gas leakage detection and auto shut-off system. *International Journal on Advanced Computer Engineering and Communication Technology*, 14(1).
- [4] Attallah, O. (2023). Multitask deep learning-based pipeline for gas leakage detection via e-nose and thermal imaging multimodal fusion. *Chemosensors*, 11, 364. <https://doi.org/10.3390/chemosensors11070364>
- [5] Dlodlo, N., & Kalezhi, J. (2015). The Internet of Things in agriculture for sustainable rural development. *International Journal of Distributed Sensor Networks*, 11(5), 1–12. <https://doi.org/10.1155/2015/530187>
- [6] Gubbi, J., Buyya, R., Marusic, S., & Palaniswami, M. (2013). Internet of Things (IoT): A vision, architectural elements, and future directions. *Future Generation Computer Systems*, 29(7), 1645–1660. <https://doi.org/10.1016/j.future.2013.01.010>
- [7] Han, S., Xu, X., & Wang, L. (2021). Real-time gas leakage detection system using wireless sensor networks. *Journal of Loss Prevention in the Process Industries*, 70, 104425. <https://doi.org/10.1016/j.jlp.2021.104425>
- [8] Liu, Z., Wang, Y., & Zhang, L. (2019). Design of an IoT-based gas leakage monitoring and alarm system for smart homes. *Sensors*, 19(9), 2036. <https://doi.org/10.3390/s19092036>
- [9] Shinde, P., & Kulkarni, P. (2020). Internet of Things based LPG gas leakage detection and alert system. *Procedia Computer Science*, 171, 264–271. <https://doi.org/10.1016/j.procs.2020.04.028>
- [10] Yoon, S., Kim, J., & Lee, H. (2018). A smart gas monitoring system based on wireless sensor networks. *IEEE Sensors Journal*, 18(20), 8425–8432. <https://doi.org/10.1109/JSEN.2018.2864532>